

Arizona Flash Flood Inundation AI Early Warning System (AFFI)

A Physics-Guided Artificial Intelligence Framework for
Real-Time Flood Inundation Forecasting in Rural Southern Arizona

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Prepared by: Solman Raju Sarva, MSc — AI & Hydrology Researcher and William D. O'Brien, PE, CFM — Principal Engineer, NextGen Engineering Inc.

Pilot Watershed: Upper Sonoita Creek Watershed, near Patagonia, Santa Cruz County, AZ

SCOPE: This “white paper” presents a technical framework piloted on the **Upper Sonoita Creek Watershed (HUC 150503010204)** in Santa Cruz County, Arizona. The architecture, methodology, and deliverables described herein are explicitly designed to be **transferable to any watershed in Arizona or the United States** with minimal reconfiguration. This “white paper” is to build collaboration for the development of AFFI early warning systems.

1. Executive Summary

Flash floods are the deadliest natural hazard in Arizona, killing an average of 14 people per year and causing \$180 million in annual property damage. The fundamental problem is not the flood — it is the **time gap in warning**. Current physics-based flood simulation tools require 4 to 6 hours to produce a flood inundation map. A monsoon flash flood in Southern Arizona arrives in under 2 hours. That gap between warning and flood impact is where lives are lost.

The **Arizona Flash Flood Inundation AI Early Warning System (AFFI)** closes this gap. AFFI is a three-stage, physics-guided artificial intelligence pipeline that ingests real-time weather forecast data and delivers a probabilistic flood inundation map to authorized government personnel at Emergency Operations Centers (EOC) in **about 60 seconds** — 360 times faster than current methods — with accuracy comparable to physics-based models.

This document describes the complete technical architecture of AFFI, piloted on the **Upper Sonoita Creek Watershed (HUC 150503010204)** in Patagonia, Santa Cruz County, Arizona. Every component of this system is designed for **generalizability**: the same pipeline, with watershed-specific data substitution, can be deployed for any watershed in Arizona or the United States.

The development of the system is broken into **six tasks**, each with specific, verifiable deliverables. All data sources are publicly available from federal agencies at no licensing cost. Prototype work on Task 1 has already been completed and validated. The flood plain modeling software, Hec-Ras, is also publicly available.

Metric	Current Capability	AFFI	Improvement
Time to flood map	4–6 hours (HEC-RAS)	< 60 seconds	360× faster
Watersheds covered	Gauged only (~12% of AZ)	Any watershed in AZ	Statewide
Forecast lead time	0–2 hours	Up to 7 days	84× longer
Ensemble members	1 (deterministic)	31 (probabilistic)	Uncertainty quantified
Cost per forecast	~\$2,400 (EOC activation)	~\$0.04 (cloud compute)	60,000× cheaper
Ungauged basin coverage	Requires stream gauge calibration	Any Arizona watershed	Ungauged-ready

2. Problem Statement: Flash Flooding in Rural Southern Arizona

2.1 The Human Cost

Between 2000 and 2023, flash floods killed 287 people in Arizona — more than tornadoes, hurricanes, and earthquakes combined in the same period ([NOAA Storm Data, 2023](#)). The majority of fatalities occurred in vehicles, where drivers encountered flooded roadways with little or no advance warning.

The pilot area for AFFI is the Upper Sonoita Creek watershed near Patagonia in Santa Cruz County, Arizona (HUC 150503010204). This drains 55.41 mi² (143.58 km²) of semi-arid terrain and has experienced 12 major flood events since 1980, with peak flows exceeding the 50-year return period on three occasions. State Route 82 near Patagonia — the only evacuation route for 3,200 residents — has been rendered impassable by flooding on multiple occasions with little or no advance warning to emergency managers.

Example: The 2006 Santa Cruz County Monsoon Flood: A single monsoon event caused \$12 million in infrastructure damage, displaced 340 residents, and closed State Route 82 for 72 hours. Emergency managers received no advance flood inundation map. Response was entirely reactive. This event is the baseline scenario AFFI is designed to mitigate.

2.2 The Infrastructure and Economic Cost

The economic cost of flash flooding in Arizona averages \$180 million annually in direct damages to roads, bridges, utilities, and structures ([FEMA, 2022](#)). Indirect costs include emergency response mobilization (\$2,400/hour for a county EOC), post-flood debris removal (\$180,000–\$450,000 per major event), long-term infrastructure repair (\$1.2M–\$8M per bridge or road segment), and agricultural losses (\$15–40M per severe monsoon season in Southern Arizona).

2.3 The Technology Gap

Southern Arizona's rural flash flood problem is uniquely difficult. The region is characterized by hundreds of ephemeral washes and arroyos — channels that are dry for 300+ days per year but can carry catastrophic flows within minutes of drenching by a monsoon rain cell. These channels are ungauged, ephemeral, braided, and peak with a rapid-response to rainfall. The time of concentration as short as 15–30 minutes for small sub-watersheds.

Current flood operational tools — NOAA's National Water Model, FEMA Flood Insurance Rate Maps, Google Flood Hub — were designed for large, gauged river systems. They provide valuable national-scale flood information but do not address the specific challenge of rural, ungauged, ephemeral flash flood prediction in Southern Arizona. **No existing operational system provides real-time, event-specific flood inundation maps for ungauged rural arroyos in Southern Arizona with lead times sufficient for emergency management action. AFFI fills this gap.**

2.4 The Climate Trend

Climate projections for the Southwest United States indicate a 15–40% increase in extreme precipitation intensity during monsoon events by 2050 ([IPCC AR6, 2021](#)). Flood events that currently occur once every 50 years will occur once every 20–30 years by mid-century. The window for investing in flood inundation early warning infrastructure — before these events become routine — is urgent.

3. Existing Systems and Their Limitations

A thorough review of existing flood inundation forecasting systems confirms that no current tool adequately addresses the rural Southern Arizona flash flood problem. The following analysis is not a criticism of these systems — each is excellent for its intended purpose. It is a demonstration that a gap exists which none of them fills.

System	Forecast Type	Lead Time	Ungauged Basins	Rural Arroyos	Real-Time Maps
AFFI (Proposed)	Probabilistic inundation maps	Up to 7 days	Yes	Yes	Yes
NOAA NWM	Streamflow forecast	Up to 7 days	Limited	No	No
Google Flood Hub	Flood forecast + maps	Up to 7 days	No	No	Partial
FEMA FIRM	Static hazard zones	None (static)	Partial	Partial	No
First Street Foundation	Risk scoring	None (static)	Partial	No	No
FLASH (RoyalHaskoningDHV)	Short-lead flood alerts	Hours to 1 day	No	No	Partial

3.1 Detailed Limitations of Each System

NOAA National Water Model (NWM): The NWM is the outstanding national-scale streamflow forecasting system. However, it requires stream gauge calibration data and is not designed for ephemeral channels. It produces streamflow forecasts, not spatial flood inundation maps. Reference: [NOAA Office of Water Prediction](#).

Google Flood Hub: Provides flood forecasting for gauged rivers globally. But does not cover ungauged basins or ephemeral arroyos. Coverage in Arizona is limited to major perennial rivers. Reference: [Google Flood Hub](#).

FEMA FIRM Maps: Static maps showing 100-year and 500-year flood zones. Not event-specific, not real-time, and not updated frequently enough to reflect current channel conditions. Reference: [FEMA Flood Map Service Center](#).

First Street Foundation: Provides property-level flood risk scores. Static, not event-specific, and not designed for emergency operations. Reference: [First Street Foundation](#).

FLASH System: Provides short-lead flash flood alerts based on radar rainfall. Does not produce spatial inundation maps and does not cover ungauged basins with sufficient resolution for rural Arizona. Reference: [NOAA Flash Flood Guidance](#).

4. Technical Solution: The AFFI Architecture

4.1 System Overview — The Correct Physical Pipeline

AFFI follows the physically correct sequence for flood prediction. This is not arbitrary — it mirrors the laws of physics governing how rainfall becomes a flood:

Rainfall Forecast → Mean Areal Watershed Precipitation → Runoff / Discharge → Flood Depth & Extent → Probability Maps → Alert

Each arrow represents a distinct physical transformation. Each transformation is handled by the most appropriate computational method — physics-based where physics is well understood, AI-based where AI has been proven to match or exceed physics-based accuracy at a fraction of the computational cost.

Stage	Input	Output	Method	Time
Stage 1: Meteorology	GFS/HRRR ensemble forecast	Mean Areal Precipitation (MAP) per watershed	Statistical downscaling	~5 sec
Stage 2: Hydrology	MAP time series + watershed attributes	Peak discharge (m ³ /s) + hydrograph	Physics-guided LSTM	~8 sec
Stage 3: Hydraulics	Peak discharge + terrain (DEM)	Flood depth + extent raster	U-Net / ResUNet	~45 sec
Stage 4: Probability	31 ensemble flood maps	Probability maps, uncertainty maps	Ensemble statistics	~2 sec
Stage 5: Benchmarking	Predicted discharge	Return period classification (2yr–100yr)	NOAA Atlas 14 comparison	~1 sec
Stage 6: Alert	All above outputs	Alert packet (JSON) + dashboard	Rule-based logic + API	~1 sec

Total pipeline time: ~ 60 seconds. This compares to 4–6 hours for the equivalent hydrology (HEC-HMS model or equal) + hydraulic (HEC- RAS model or equal) physics-based simulation modeling chain.

4.2 Why This Architecture and Not Others

After reviewing over 200 published AI flood prediction systems, this three-stage architecture is the only design that satisfies all five operational requirements simultaneously:

- **Speed:** ~ 60 seconds end-to-end — required for flash flood warning
- **Accuracy:** Comparable to physics-based models — required for government trust
- **Generalizability:** Works on ungauged watersheds — required for statewide and rural deployment
- **Interpretability:** Each stage has physical meaning — required for regulatory acceptance
- **Uncertainty quantification:** Probabilistic output — required for risk-based decisions

A single end-to-end deep learning model (rainfall directly to flood map) fails on generalizability — it memorizes the training watershed and cannot transfer. A pure physics model fails on speed. The three-stage hybrid presented is the only architecture that passes all five tests.

4.3 Stage 1: Meteorological Interface — Explained

What it does: Ingests ensemble rainfall forecasts from NOAA's Global Forecast System (GFS, 31 members, 6-

hourly updates) and the High-Resolution Rapid Refresh (HREF -High-Resolution Ensemble Forecast, ~10 members, hourly, CONUS). For each watershed, computes Mean Areal Precipitation (MAP) is the spatially averaged rainfall over the watershed extent — for forecast horizons of 1, 3, 6, 12, 24, 48, 72, and 168 hours.

Why it matters: A point rainfall measurement tells you how much rain fell at one location. A watershed responds to the average rainfall over its entire area. MAP is the physically correct input to a hydrologic model. Using point rainfall instead of MAP is a common error that leads to significant over- or under-prediction of flood response.

What is AI at this stage? Stage 1 is NOT AI. It is physics-based numerical weather prediction (NWP) from NOAA's operational models. The AI begins in Stage 2. This distinction is important for scientific integrity and regulatory acceptance.

Comparison to 10-year storm rainfall: Each MAP value is compared against NOAA Atlas 14 precipitation frequency benchmarks for Santa Cruz County, producing a return-period classification (2-year through 100-year) and an initial alert level (Advisory / Watch / Warning). This allows emergency managers to immediately understand whether an incoming storm is a routine event or a rare, high-consequence event.

4.4 Stage 2: Physics-Guided LSTM Hydrology Model — Explained

What it does: Converts the MAP time series into a predicted streamflow hydrograph (discharge over time, in m³/s) using a Long Short-Term Memory (LSTM) neural network trained on 42 years of streamflow records from 25 Arizona watersheds.

Why LSTM? Rainfall-runoff is a time-dependent process — today's soil moisture depends on yesterday's rainfall. LSTM networks are specifically designed to learn these temporal dependencies. They have been independently validated on 531 US watersheds by Kratzert et al. (2019), achieving Nash-Sutcliffe Efficiency (NSE) > 0.80 — matching or exceeding calibrated physics-based models.

HOW: LSTM is trained in two stages: base training on a high-quality anchor watershed (Walnut Gulch), followed by fine-tuning on the pilot watershed. Reference transfer learning in hydrology (Kratzert et al. 2019 already supports this)

Why "physics-guided"? The LSTM incorporates static watershed attributes (drainage area, slope, soil permeability, vegetation index, elevation, aridity index) from the USGS GAGES-II dataset. These attributes encode the physical characteristics of the watershed into the model, enabling it to predict discharge for watersheds it has never seen — the key to generalizability. Model training follows a two-stage transfer learning strategy. In Stage 1, the LSTM is trained on the Walnut Gulch Experimental Watershed — the world's most complete semi-arid flash flood dataset with 70+ years of record — learning fundamental desert hydrology: rapid infiltration, Hortonian overland flow, ephemeral channel routing, and monsoon precipitation-runoff dynamics. In Stage 2, the base model is fine-tuned on local gauge data for the target watershed (Santa Cruz at Tubac, USGS 09481740, for the Sonoita Creek pilot) using only the final layers, requiring as few as 10–15 years of local data. This approach — base training on a data-rich anchor watershed followed by targeted fine-tuning — is established in the hydrology AI literature (Kratzert et al. 2019) and is the mechanism that enables generalization to ungauged watersheds.

Comparison to 10-year storm peak runoff: The predicted peak discharge is compared against the 10-year, 25-year, 50-year, and 100-year return period flows derived from USGS regional regression equations for Arizona. This tells emergency managers "how rare or how extreme" this event is.

4.5 Stage 3: U-Net Hydraulic Surrogate Model — Explained

What it does: Converts predicted discharge into a spatial flood inundation map (flood depth and extent raster)

using a U-Net convolutional neural network trained on 250 physics-based HEC-RAS flood simulations.

Why U-Net? U-Net was originally developed for medical image segmentation and is now the standard architecture for spatial prediction problems. It takes a 2D input (terrain + discharge) and produces a 2D output (flood depth map) — exactly the transformation needed for hydraulic modeling.

Why train on HEC-RAS? HEC-RAS is the industry-standard open-source hydraulic model used by FEMA and USACE. By training the U-Net on HEC-RAS outputs, AFFI inherits the physical accuracy of HEC-RAS while operating 360× faster. The U-Net is a surrogate (emulator) of HEC-RAS, not a replacement.

Physics constraints enforced: Three physics constraints are applied to the U-Net output: (1) Gravity constraint — water cannot flow uphill; (2) Monotonicity constraint — higher discharge must produce equal or greater flood inundation area and extent; (3) Mass conservation — total predicted runoff volume cannot exceed total rainfall volume. These constraints ensure physical plausibility even for scenarios outside the training distribution.

Comparison to 10-year to 100-year flood inundation: The predicted flood map is compared against pre-computed HEC-RAS flood maps for the 10-year, 25-year, 50-year, and 100-year return period events. This provides emergency managers with a direct, intuitive benchmark: "This event is expected to produce flooding comparable to a 25-year storm."

4.6 Stage 4: Probabilistic Output

The full three-stage pipeline is executed independently for each of the 31 GFS ensemble members, producing 31 flood maps for each forecast event. These maps are stacked and 4 critical flood maps are produced. This ensemble approach quantifies forecast uncertainty and produces the following 6 output products:

Output Product	Description	Primary User
Flood Inundation Maps		
Median flood map	Best-estimate flood depth and extent	EOC Staff
90th percentile map	Worst-case scenario (10% chance of exceeding)	EOC staff, Public Works Director
Probability map	P(depth > 0.5m) for each pixel, 0–100%	EOC Staff
Uncertainty map	Standard deviation of depth across 31 maps	Technical reviewer
Warnings		
Alert classification	Advisory / Watch / Warning vs. return period	County OEM
Time-to-peak estimate	Hours until maximum flooding at each location	Road closure coordinator

4.7 Alert Policy

Critical Policy Decision: All AFFI alerts are delivered exclusively to authorized government personnel — county emergency managers, public works directors, and EOC staff. No alerts are broadcast directly to the public. This design eliminates the liability risk of automated public warnings and ensures that all public communications are reviewed and authorized by responsible government officials.

4.8 Lead-Time-Dependent Confidence Framework

Lead Time	Primary Data Source	Confidence Level	Primary Use
2–7 days	GFS ensemble (31 members)	Low–Medium	Pre-positioning resources, early briefings
6–24 hours	HRRR + updated GFS	Medium–High	Road closure decisions, evacuation planning
0–6 hours	MRMS radar + nowcast	High	Active road closures, emergency response

5. Task Breakdown with Deliverables

The development of AFFI is organized into **seven tasks across four deployment stages**. Tasks are staged deliberately — Stage 1 establishes the AI foundation on a high-quality anchor watershed; Stage 2 adapts the system to the pilot watershed; Stage 3 delivers the full operational pipeline; Stage 4 scales statewide.". Each task has a defined scope, a step-by-step technical description, and specific, verifiable deliverables. Tasks are sequential but partially overlapping. No completion dates are specified — tasks are complete when deliverables meet the stated acceptance criteria.

Stage	Watershed	Tasks	Purpose	Dependency
Stage 1	Walnut Gulch, AZ	T1 + T2a	Teach AI semi-arid desert hydrology using pristine anchor dataset	Foundation
Stage 2	Sonoita Creek / Santa Cruz	T2b + T3	Fine-tune LSTM on local gauge; build hydraulic surrogate	After T2a
Stage 3	Sonoita Creek	T4 + T5 + T6	Full probabilistic system: risk maps, benchmarks, alerts	After T3
Stage 4	Other AZ watersheds	T7	Scale operational system statewide using transfer learning	After T6

Task 1: Meteorological Forecast Interface

Objective: Build and validate the system that ingests real-time weather forecast data and computes Mean Areal Precipitation (MAP) for the Upper Sonoita Creek Watershed, with alert threshold logic benchmarked against NOAA Atlas 14 return periods.

Step-by-Step:

- **Step 1.1 — Define Area of Interest (AOI):** Load the Upper Sonoita Creek Watershed boundary (HUC 150503010204) from the USGS National Map. Define bounding box (N=31.85, S=31.47, E=-110.50, W=-110.90). This is the spatial domain for all precipitation calculations.
- **Step 1.2 — Ingest GFS Ensemble Forecast:** Connect to the Open-Meteo API (free, no key required) to retrieve 31-member GFS ensemble precipitation forecasts at 6-hourly intervals for 7-day lead times. Parse and store as structured arrays.
- **Step 1.3 — Compute Mean Areal Precipitation (MAP):** Spatially average the gridded GFS precipitation

over the watershed boundary using area-weighted averaging. Compute MAP for accumulation windows of 1, 3, 6, 12, 24, 48, 72, and 168 hours.

- **Step 1.4 — Load NOAA Atlas 14 Benchmarks:** Load precipitation frequency estimates for Santa Cruz County, AZ from NOAA Atlas 14 Volume 1 for return periods of 2, 5, 10, 25, 50, and 100 years at all accumulation durations. These are the physical benchmarks against which all forecast rainfall is compared.
- **Step 1.5 — Compute Return Period Classification:** For each MAP value and accumulation duration, determine which return period bracket it falls into (e.g., "between 10-year and 25-year"). Compute a Storm Severity Index ($SSI = MAP / MAP_{10yr}$) for each duration.
- **Step 1.6 — Apply Alert Threshold Logic:** Apply the following thresholds calibrated for the Sonoita Creek watershed: Advisory (0.25"/1hr, 0.75"/24hr), Watch (0.50"/1hr, 1.25"/24hr), Warning (1.00"/1hr, 2.00"/24hr). Generate an alert level (GREEN / ADVISORY / WATCH / WARNING) for each ensemble member and each lead time.
- **Step 1.7 — Generate Alert Packet:** Compile all outputs into a structured JSON alert packet containing: alert level, MAP values, return period classifications, ensemble statistics (mean, 10th, 50th, 90th percentile), and time-to-peak estimate.
- **Step 1.8 — Dummy Rainfall Stress Test:** Validate the alert logic using synthetic rainfall scenarios spanning the full range from light shower to extreme cloudburst. Confirm that each scenario triggers the correct alert level and return period classification.

D1.1 — Validated Python module: `task1_met_interface.py` — ingests GFS ensemble, computes MAP, classifies return period, generates alert level.

D1.2 — Alert Packet JSON: `task1_alert_packet.json` — structured output for all 31 ensemble members, all lead times, all accumulation durations.

D1.3 — Forecast Dashboard: `task1_forecast_dashboard.png` — 4-panel visualization: MAP time series, ensemble spread, return period comparison, alert level timeline.

D1.4 — Dummy Stress Test Report: `task1_dummy_benchmark.png` — 2-panel chart showing all synthetic scenarios plotted against 2yr–100yr NOAA Atlas 14 benchmarks with alert-level coloring.

D1.5 — Technical Memo: Written description of Task 1 methodology, data sources, threshold calibration rationale, and validation results. Suitable for inclusion in a peer-reviewed methods section.

Status: COMPLETE — All D1.1 through D1.4 deliverables have been produced and validated for the Upper Sonoita Creek Watershed.

Task 2a: Base LSTM Training — Walnut Gulch Anchor Watershed (Stage 1)

Objective: Train and validate the base LSTM on the Walnut Gulch Experimental Watershed (USDA ARS) — the highest-quality, longest-record semi-arid watershed dataset in the United States — establishing a physics-informed rainfall-to-discharge model that serves as the foundation for fine-tuning on any Arizona watershed.

Step-by-Step:

- **Step 2a.1 — Collect Historical Streamflow Data:** Download the Walnut Gulch Experimental Watershed dataset from the USDA Agricultural Research Service (ARS) Southwest Watershed Research Center. This dataset contains 70+ years of sub-hourly rainfall and runoff records from 89 rainfall gauges and 12 flumes across a 150 km² semi-arid watershed near Tombstone, AZ. This is the anchor training dataset — chosen because it is the world's most complete long-term semi-arid flash flood dataset, with no missing monsoon seasons.
- **Step 2a.2 — Collect Historical Rainfall Data:** Download ERA5 reanalysis precipitation via the Open-Meteo Archive API (free, no API key required) at hourly resolution, 0.25° grid, covering 1940–present. Extract 9 grid points covering the Walnut Gulch domain. ERA5 is used as the primary gridded precipitation input because it provides a physically consistent, bias-corrected reanalysis that matches the temporal resolution required by the LSTM. Note: ERA5's 0.25° resolution may underrepresent small intense monsoon cells; the LSTM is trained to learn this systematic bias as part of the rainfall-runoff relationship.
- **Step 2a.3 — Compile Watershed Attributes:** Extract 6 static attributes for each watershed from USGS GAGES-II dataset: drainage area (km²), mean slope (%), soil permeability (cm/hr), NDVI vegetation index, mean elevation (m), and aridity index. For Walnut Gulch, these attributes are available at high precision from the USDA ARS documentation and USGS GAGES-II dataset. These same 6 attributes will be used for all future fine-tuning watersheds, enabling transfer learning.
- **Step 2a.4 — Data Quality Control:** Remove periods with missing data, sensor malfunctions, or known rating curve errors. Apply gap-filling for short gaps (<7 days) using linear interpolation. Document all data quality decisions.
- **Step 2a.5 — Design LSTM Architecture:** Implement a single-watershed LSTM for Walnut Gulch with: 30-day lookback window for dynamic inputs (hourly rainfall, temperature, antecedent soil moisture proxy, day-of-year); static attribute embedding layer encoding 6 watershed attributes; 2 LSTM layers (128 hidden units each); dropout regularization (0.3); output layer predicting hourly discharge (m³/s). Note: The lookback window is 30 days, not 365 days, because Walnut Gulch is an ephemeral watershed — antecedent conditions reset rapidly between monsoon events. This is physically appropriate for semi-arid systems.
- **Step 2a.6 — Training Split:** Train on Walnut Gulch data from 1980–2010 (30 years, ~60 monsoon seasons), validate on 2011–2018 (8 years), test on 2019–2024 (5 years, completely held out). This temporal split ensures the model is tested on the most recent monsoon behavior, including any

climate-trend effects. Use Adam optimizer, MSE loss on log-transformed discharge (to give equal weight to low flows and peak events), batch size 128, 100 epochs with early stopping.

- **Step 2a.7 — Validate Against Physics-Based Baseline:** Compare LSTM performance against: (1) a calibrated USDA ARS rainfall-runoff model for Walnut Gulch, which serves as the physics baseline; (2) a simple linear regression benchmark. Report NSE, KGE, PBIAS, and peak flow timing error (hours). Target: NSE > 0.75 on validation years; NSE > 0.65 on held-out test years.

Step 2.8 — Return Period Discharge Estimation: For each watershed, derive the 2yr, 10yr, 25yr, 50yr, and 100yr return period discharge values using USGS regional regression equations for Arizona. These become the hydraulic benchmarks for Task 3.

D2a.1 — Walnut Gulch training dataset: hourly ERA5 precipitation + USDA ARS discharge, 1980–2024, quality-controlled, documented.

D2a.2 — Trained base LSTM model: `lstm_base_walnutgulch.pt` — PyTorch model, documented architecture, training history, hyperparameters.

D2a.3 — Validation report: NSE, KGE, PBIAS for train/validate/test splits. NSE > 0.75 (validation), NSE > 0.65 (test).

D2a.4 — Hydrograph comparison figure: Predicted vs. observed discharge for 5 major Walnut Gulch flood events including peak flow timing.

D2a.5 — Transfer-readiness checklist: Documented proof that the base model generalizes — tested on one held-out sub-watershed within Walnut Gulch.

Task 2b: LSTM Fine-Tuning — Sonoita Creek / Santa Cruz Pilot Watershed (Stage 2)

Objective: Fine-tune the Walnut Gulch base LSTM (Task 2a) on the Santa Cruz River at Tubac gauge (USGS 09481740) — the best available long-record gauge for the Upper Sonoita Creek pilot area — using transfer learning to adapt the base model to local semi-arid hydrology with only 23 years of local data.

Step-by-Step:

- **Step 2b.1 — Data Download:** USGS gauge 09481740 (Santa Cruz at Tubac), 15-minute interval, 1995–present, ~30 years, ~1.05 million records. Note: Original Sonoita Creek gauge 09481500 was decommissioned in 1972 and cannot be used.
- **Step 2b.2 — Precipitation Input:** ERA5 via Open-Meteo Archive API, 9 grid points covering the Upper Sonoita Creek bounding box (N=31.85, S=31.47, E=-110.50, W=-110.90), hourly, 2002–present.
- **Step 2b.3 — Transfer Learning:** Load the Walnut Gulch base model weights. Freeze the LSTM encoder layers. Fine-tune only the output layer and static attribute embedding on Santa Cruz at Tubac data (2002–2020 train, 2021–2025 test). This requires only 18 years of local data — a fraction of what a from-scratch training would need.
- **Step 2b.4 — Monsoon Event Validation:** Validate on the August 2014 Hurricane Odile remnant event (observed peak: ~6,520 cfs at Tubac) and 3 other documented Santa Cruz flood events.

D2b.1 — Santa Cruz at Tubac dataset: hourly ERA5 + 15-min USGS discharge, 2002–2025, quality-controlled.

D2b.2 — Fine-tuned LSTM: `lstm_finetune_sonoita.pt` — PyTorch model demonstrating transfer from Walnut Gulch.

D2b.3 — Transfer learning validation: NSE comparison of base model (no fine-tuning) vs. fine-tuned model on Santa Cruz test years. Shows value of transfer learning.

D2b.4 — August 2014 Odile event replay: Hydrograph of predicted vs. observed for the highest-discharge event on record at Tubac.

Task 3: U-Net Hydraulic Surrogate Model

Objective: Train and validate a U-Net convolutional neural network that converts predicted discharge into spatial flood inundation maps, using HEC-RAS simulations as training data.

Step-by-Step:

- **Step 3.1 — Prepare Terrain Data:** Download USGS 3DEP 1-meter lidar DEM for the Upper Sonoita Creek Watershed (Stage 2 pilot). The 25-watershed expansion is a Stage 4 activity (Task 7). At Stage 2, the U-Net is trained exclusively on the pilot watershed using HEC-RAS simulations

specific to Sonoita Creek terrain.

- **Step 3.2 — Build HEC-RAS Models:** Construct a 2D HEC-RAS hydraulic model for the Upper Sonoita Creek Watershed using the conditioned 1-meter lidar DEM and Manning's roughness values from NLCD land cover. Calibrate against available historical high-water marks from the 2006 Santa Cruz County monsoon flood (the baseline event defined in Section 2.1) and the 2014 Odile remnant event.
- **Step 3.3 — Run HEC-RAS Simulations:** Execute 50 HEC-RAS simulations for the Upper Sonoita Creek Watershed (10 discharge levels: Q2, Q5, Q10, Q25, Q50, Q100, and 4 intermediate levels; 5 replicates with varied Manning's n to capture roughness uncertainty). Expanding to 25 watersheds × 250 simulations is a Stage 4 activity. Starting with 50 high-quality simulations on one watershed is sufficient to train the U-Net surrogate for Stage 2 operational deployment.
- **Step 3.4 — Design U-Net Architecture:** Implement a ResUNet (U-Net with residual connections) with: input channels = DEM + slope + channel distance + discharge value; output channel = flood depth (meters); 4 encoder/decoder levels; skip connections at each level.
- **Step 3.5 — Train U-Net:** Train on 200 simulations (20 watersheds × 10 discharge levels), validate on 25 simulations (5 watersheds × 5 discharge levels), test on 25 simulations (5 held-out watersheds × 5 discharge levels). Use Adam optimizer, combined MSE + structural similarity loss.
- **Step 3.6 — Enforce Physics Constraints:** Post-process U-Net output to enforce: (1) gravity constraint — zero depth for pixels above predicted water surface elevation; (2) monotonicity — higher discharge = equal or greater extent; (3) mass conservation — total volume check.
- **Step 3.7 — Validate Against HEC-RAS:** Compare U-Net flood maps against HEC-RAS maps on held-out watersheds using Intersection over Union (IoU), Critical Success Index (CSI), and mean absolute depth error (MAE). Target: IoU > 0.75.
- **Step 3.8 — Generate Return Period Flood Map Library:** For each watershed, generate pre-computed flood maps for Q2, Q10, Q25, Q50, and Q100 using both HEC-RAS (reference) and U-Net (operational). These become the benchmark flood maps for comparison in the alert system.

D3.1 — Hydraulically conditioned DEMs for all 25 watersheds, documented processing workflow.

D3.2 — 250 HEC-RAS flood simulation outputs (depth rasters), organized by watershed and discharge level. This is the primary training dataset for the U-Net.

D3.3 — Trained U-Net model: `UNET_hydraulic_model.pt` — PyTorch model file with documented architecture and training history.

D3.4 — Validation report: IoU, CSI, MAE for all 25 watersheds. IoU > 0.75 on held-out watersheds.

D3.5 — Return period flood map library: Pre-computed flood maps (GeoTIFF) for Q2, Q10, Q25, Q50, Q100 for all 25 watersheds — both HEC-RAS reference and U-Net operational versions.

Task 4: Probabilistic Risk Products

Objective: Execute the full three-stage pipeline for all 31 GFS ensemble members and generate probabilistic flood risk products suitable for emergency management decision-making.

Step-by-Step:

- **Step 4.1 — Ensemble Pipeline Execution:** Run the complete Task 1 → Task 2 → Task 3 pipeline independently for each of the 31 GFS ensemble members. This produces 31 flood depth maps for each forecast event.

- **Step 4.2 — Compute Ensemble Statistics:** For each pixel in the flood map domain, compute: median depth (50th percentile), 90th percentile depth (worst-case), probability of exceeding 0.5m depth, standard deviation (uncertainty), and ensemble agreement index.
- **Step 4.3 — Generate Probability Map:** Create a raster where each pixel value represents the probability (0–100%) that flood depth will exceed 0.5 meters. This is the primary decision-support product for emergency managers.
- **Step 4.4 — Generate Uncertainty Map:** Create a raster showing the standard deviation of predicted depth across 31 ensemble members. High uncertainty areas require more conservative decision-making.
- **Step 4.5 — Compute Time-to-Peak:** For each ensemble member, identify the time step at which peak discharge occurs at the watershed outlet. Report the ensemble mean and spread of time-to-peak as a lead time estimate for emergency managers.
- **Step 4.6 — Generate EOC Dashboard:** Compile all probabilistic products into a multi-panel dashboard suitable for display in a county Emergency Operations Center: median map, probability map, uncertainty map, ensemble hydrograph, alert timeline, and return period comparison.

D4.1 — Probabilistic flood map set: Median, 90th percentile, probability, and uncertainty rasters (GeoTIFF) for each forecast event.

D4.2 — EOC Dashboard: `task4_eoc_dashboard.png` — 6-panel visualization suitable for emergency operations center display.

D4.3 — Ensemble hydrograph: Time series of predicted discharge for all 31 ensemble members, with median and uncertainty bounds highlighted.

D4.4 — Time-to-peak report: Ensemble mean and spread of time-to-peak for each forecast event, formatted for emergency manager briefings.

Task 5: 10-Year Storm Benchmarking and Return Period Comparison

Objective: Establish a rigorous comparison framework that allows emergency managers to understand any predicted flood event in terms of its rarity and severity relative to standard engineering design storms (2-year through 100-year return periods).

Step-by-Step:

- **Step 5.1 — Define Return Period Benchmarks:** For the Upper Sonoita Creek Watershed, compile the following benchmarks from authoritative federal sources: (a) NOAA Atlas 14 precipitation frequency estimates for 2yr, 5yr, 10yr, 25yr, 50yr, 100yr at 1hr, 3hr, 6hr, 24hr durations; (b) USGS regional regression equations for peak discharge at the same return periods; (c) Pre-computed HEC-RAS flood maps for each return period (from Task 3, D3.5).
- **Step 5.2 — Build Comparison Engine:** Implement a comparison module that takes any predicted rainfall, discharge, or flood map and automatically determines: which return period bracket it falls into; how it compares to the 10-year design storm (the standard engineering benchmark); what percentage of the 100- year floodplain is inundated.
- **Step 5.3 — Historical Event Validation:** Apply the full pipeline to 5 major historical Arizona flood events (1983, 1993, 2006, 2014, 2017). Compare predicted flood maps against post-event surveys, FEMA damage assessments, and USGS peak flow records. Report IoU, CSI, and return period classification accuracy.
- **Step 5.4 — Generate Benchmark Comparison Report:** For each historical event, produce a

side-by-side comparison: AFFI predicted map vs. post-event survey vs. FEMA FIRM 100-year zone. Quantify the improvement in spatial accuracy over static FIRM maps.

- **Step 5.5 — Sensitivity Analysis:** Test the pipeline's sensitivity to input uncertainty by varying: GFS forecast error ($\pm 20\%$), watershed attribute uncertainty ($\pm 10\%$), DEM resolution (1m vs. 10m). Report how these uncertainties propagate to flood map uncertainty.

D5.1 — Return period benchmark database: NOAA Atlas 14 precipitation + USGS discharge + HEC-RAS flood maps for 2yr, 5yr, 10yr, 25yr, 50yr, 100yr for all 25 watersheds.

D5.2 — Historical event validation report: AFFI performance on 5 major AZ flood events. IoU > 0.75 vs. post-event surveys.

D5.3 — Benchmark comparison figures: Side-by-side maps (AFFI vs. post-event survey vs. FEMA FIRM) for all 5 historical events.

D5.4 — Sensitivity analysis report: Quantified uncertainty propagation from input errors to flood map errors.

D5.5 — Return period classification accuracy table: For all 5 historical events, report whether AFFI correctly classified the event's return period within one bracket (e.g., predicted "25–50 year" for an actual 35-year event).

Task 6: Alert System, Government API, and Decision Dashboard

Objective: Integrate all pipeline components into an operational alert system with a secure government API and an EOC-ready decision dashboard, with alert logic defined and controlled by county emergency management agencies.

Step-by-Step:

- **Step 6.1 — Define Alert Logic:** In collaboration with Santa Cruz County OEM, define the alert threshold matrix: which combination of return period classification, probability map values, and time-to-peak triggers each alert level (Advisory / Watch / Warning). Document all decision rules in a formal Alert Logic Specification document.
- **Step 6.2 — Build Alert Engine:** Implement the alert engine as a Python module that: reads the Task 4 probabilistic outputs; applies the Task 6.1 alert logic; generates a structured JSON alert packet; logs all decisions with timestamps and input values for audit purposes.
- **Step 6.3 — Build Government API:** Develop a RESTful API (FastAPI framework) that: accepts watershed ID and forecast timestamp as inputs; executes the full AFFI pipeline; returns the alert packet and flood map URLs; requires API key authentication for all government users.
- **Step 6.4 — Build EOC Dashboard:** Develop a web-based dashboard (Streamlit or Dash framework) that: displays the current alert level with color coding; shows the median and 90th percentile flood maps on an interactive map; displays the ensemble hydrograph with uncertainty bounds; shows the return period comparison table; updates automatically every 6 hours with new GFS data.
- **Step 6.5 — Operational Testing:** Deploy the system for live testing during the Arizona monsoon season. Log all forecast events, alert levels, and (where available) observed outcomes. Compute false alarm rate, missed event rate, and lead time accuracy.
- **Step 6.6 — Documentation and Training:** Produce complete system documentation: API reference guide, dashboard user manual, alert logic specification, data flow diagram, and maintenance procedures. Conduct training sessions with Santa Cruz County OEM staff.

D6.1 — Alert Logic Specification document: Formal definition of all alert thresholds, decision rules, and override procedures. Signed off by Santa Cruz County OEM.

D6.2 — Alert engine module: `task6_alert_engine.py` — validated Python module with full audit logging.

D6.3 — Government API: Deployed FastAPI application with authentication, documentation, and test suite. API reference guide included.

D6.4 — EOC Dashboard: Deployed Streamlit/Dash application with user manual. Demonstrated to Santa Cruz County OEM.

D6.5 — Operational test report: Performance metrics from live monsoon season testing: false alarm rate, missed event rate, lead time accuracy.

D6.6 — Complete system documentation package: API guide, user manual, alert logic spec, data flow diagram, maintenance procedures, training materials.

Task 7: Statewide Scaling — Transfer to Additional Arizona Watersheds (Stage 4)

Objective: Deploy the validated AFFI pipeline to additional Arizona watersheds using the transfer learning framework established in Tasks 2a and 2b, demonstrating that the marginal cost of adding a new watershed is 40–80 hours of engineering effort, not a full system rebuild.

Key steps:

- **Step 7.1** — Watershed Selection: Prioritize AZ watersheds using 4 criteria: (a) USGS 3DEP 1-meter lidar available, (b) NOAA Atlas 14 county data available, (c) flash flood risk to populated area, (d) no existing operational flood warning system.
- **Step 7.2** — Rapid Deployment Protocol: For each new watershed, execute the 8-component checklist from Section 8 (Scalability). Target: full deployment in 40–80 hours per watershed.
- **Step 7.3** — Cross-Watershed Validation: Test whether the Sonoita Creek fine-tuned LSTM (Task 2b) transfers to a new watershed without retraining. Report NSE degradation vs. full fine-tuning. This directly tests generalizability.
- **Step 7.4** — Statewide Dashboard: Extend the Task 6 EOC Dashboard to display alert levels for all deployed watersheds on a single Arizona-wide map.

D7.1 — Rapid deployment checklist: Documented per-watershed configuration steps, verified on 3 pilot Arizona watersheds beyond Sonoita Creek.

D7.2 — Cross-watershed generalization report: NSE, KGE on 3 new watersheds with and without fine-tuning.

D7.3 — Statewide alert dashboard: Extended Streamlit/Dash application showing all deployed watersheds.

D7.4 — Deployment cost audit: Actual engineering hours per watershed for the 3 pilot expansions, validating the 40–80 hour estimate from Section 8.

6. Data Sources and Technical Feasibility

All data required for AFFI development and operation are publicly available from federal agencies at no licensing cost. This eliminates a major category of project risk and ensures long-term operational sustainability.

Data Type	Source	Resolution	Coverage	Cost
Streamflow records(anchor)	USDA ARS — Walnut Gulch Experimental Watershed	Sub-hourly	0+ years, 89 rain gauges, 12 flumes	Free
Historical precipitation	ERA5 via Open-Meteo Archive API	0.25°, hourly	940–present, global	Free
Streamflow (pilot watershed)	USGS NWIS — Gauge 09481740 (Santa Cruz at Tubac)	15-min interval	1995–present, 30yr record	Free
Terrain (DEM)	USGS 3DEP	1-meter lidar	Statewide AZ	Free
Real-time forecast	NOAA GFS	~13km, 6-hourly	31 ensemble members	Free
High-res forecast	NOAA HRRR	3km, hourly	18 ensemble members	Free
Radar rainfall	NOAA MRMS	1km, 2-min	Real-time, CONUS	Free
Soil data	USDA SSURGO	1:24,000	Statewide AZ	Free
Land cover	USGS NLCD	30m	Statewide AZ	Free
Watershed attributes	USGS GAGES-II	Per watershed	9,322 US watersheds	Free
Flood benchmarks	NOAA Atlas 14	Per county	All AZ counties	Free

6.1 Existing Progress — Technical Feasibility

Significant foundational work has already been completed as of April 2026, demonstrating technical feasibility, capability and commitment:

- 6.1.1 A prototype meteorological forecast interface has been developed and tested for the Sonoita Creek watershed (HUC 150503010204), successfully ingesting GFS ensemble data and computing mean areal precipitation in real time.
- 6.1.2 Alert threshold logic has been calibrated against NOAA Atlas 14 precipitation frequency data for Santa Cruz County, with Advisory / Watch / Warning levels benchmarked against 10-year, 25-year, and 50-year return periods.
- 6.1.3 The complete data pipeline architecture has been designed, documented, and partially implemented.
- 6.1.4 A dummy rainfall stress-test framework has been developed and validated against NOAA Atlas 14 IDF benchmarks for all return periods from 2-year through 100-year.
- 6.1.5 Task 1 deliverables D1.1 through D1.4 are complete and validated.

7 Validation Plan and Performance Targets

AFFI will be validated against four independent benchmarks before operational deployment. Specific, measurable performance targets are stated below. These targets are based on published standards in the peer-reviewed hydrology literature.

Validation Level	Method	Target	Reference Standard
In-sample accuracy	LSTM tested on held-out years (2001–2006) from 20 training watersheds	NSE > 0.75, KGE > 0.70	Moriassi et al. 2007
Out-of-sample generalization	LSTM tested on 5 completely held-out watersheds never seen in training	NSE > 0.65	Kratzert et al. 2019
Historical event validation	Full pipeline tested on 5 major AZ flood events (1983, 1993, 2006, 2014, 2017)	IoU > 0.75 vs. post-event surveys	FEMA post-disaster assessments
Speed benchmark	Wall-clock time from GFS input to flood map output	< 60 seconds	Flash flood warning lead time requirement

7.1 Baseline Comparisons

AFFI will be compared against three baselines to demonstrate its contribution:

- 7.1.1 **Baseline 1 — Simple LSTM (no watershed attributes):** Proves that physics-guided design adds value over generic AI
- 7.1.2 **Baseline 2 — HEC-HMS + HEC-RAS (full physics):** Proves that AFFI approaches physics-model accuracy at 360× the speed
- 7.1.3 **Baseline 3 — FEMA static FIRM maps:** Proves that dynamic event-based maps outperform static hazard maps for real-time decisions

8 Scalability to Any Watershed

Design Principle: Every component of AFFI was designed from the ground up to be transferable to any watershed in Arizona or the United States. The Upper Sonoita Creek Watershed is the pilot — not the only watershed that could be used. The architecture is generalizable by design, not by accident.

To deploy AFFI for a new watershed, the following watershed-specific inputs must be substituted. All other components (LSTM architecture, U-Net architecture, alert logic framework, API, dashboard) remain unchanged:

Component	What Changes Per Watershed	Data Source	Estimated Effort
AOI / Bounding Box	Watershed boundary polygon and bounding box coordinates	USGS National Map	1 hour
NOAA Atlas 14 Benchmarks	County-specific precipitation frequency estimates	NOAA Atlas 14	2 hours
Alert Thresholds	Advisory/Watch/Warning rainfall thresholds calibrated to local conditions	County OEM + NOAA Atlas 14	4 hours
DEM	1-meter lidar DEM for new watershed domain	USGS 3DEP	4 hours
HEC-RAS Model	New 2D hydraulic model for new watershed	DEM + NLCD + field survey	20–40 hours
LSTM Fine-Tuning	Optional: fine-tune on local gauge data if available	USGS NWIS	8 hours
U-Net Re-Training	Add new HEC-RAS simulations to training set; retrain U-Net	HEC-RAS outputs from above	10 hours

Estimated total effort per new watershed: 40–80 hours, depending on whether local gauge data is available for LSTM fine-tuning. This is the marginal cost of statewide expansion — not the full system development cost.

Watersheds where AFFI is immediately applicable without modification: Any Arizona watershed with an available USGS 3DEP lidar DEM and NOAA Atlas 14 county data. For watersheds outside Arizona, the only additional requirement is downloading the relevant NOAA Atlas 14 volume for the applicable state.

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